

Companion to Bird & de Moor, Algebra of Programming

§5. Datatypes in Allegories

§5.1. Relators

(5.1a) A relator $F : \mathcal{C} \rightarrow \mathcal{D}$ is a 2-functor between allegories: it preserves 1 , composition and $\sqsubseteq$.

	the statement	(5.1b)
1	$\text{Map}(f) \implies \text{Map}(F(f))$ and $\text{Map}(f) \implies F(f^\circ) = F(f)^\circ$. Lemma 5.1	
2	Over a tabular allegory a functor is a relator $\iff$ it preserves $^\circ$. Theorem 5.1 the book’s reverse direction (preserving $^\circ \implies$ relator) needs the source to be Rel : the counterexample over a tabular allegory that is not Rel shows it can fail	
3	$F(R^\circ) = F(R)^\circ$ for every R , so $F(R)^\circ$ needs no bracket.	
4	Two relators agreeing on maps are equal. Corollary 5.1	
5	$\text{Coreflexive}(X) \wedge \text{Coreflexive}(Y) \implies F(X \cap Y) = F(X) \cap F(Y)$ Ex 5.2	
6	$F(R \cap S) \sqsubseteq F(R) \cap F(S)$, and strictly. Ex 5.2	
7	$F(\text{dom}(R)) = \text{dom}(F(R))$ Ex 5.5	

The **power relator** $P \multimap xs \ P(R) \ ys \iff (\forall a \in xs. \ \exists b \in ys. \ a \ R \ b) \wedge (\forall b \in ys. \ \exists a \in xs. \ a \ R \ b)$ — is where the fourth is strict: for $R = \{(a_1, b_1), (a_2, b_2)\}$ and $S = \{(a_1, b_2), (a_2, b_1)\}$ the pair $(\{a_1, a_2\}, \{b_1, b_2\})$ is in $P(R) \cap P(S)$, while $R \cap S = \emptyset$.

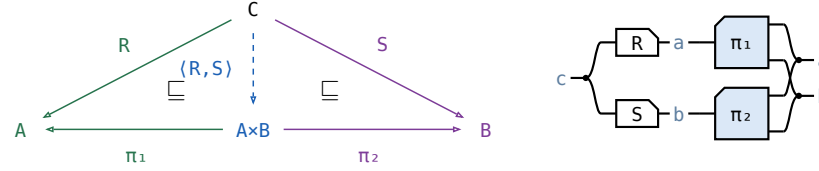
§5.2. Relational products

$$\langle R, S \rangle \triangleq (R \pi_1^\circ) \cap (S \pi_2^\circ) \text{ — the fork of } R : C \rightarrow A \text{ and } S : C \rightarrow B, \text{ with } (\pi_1, \pi_2) \text{ the tabulation of } \tau. \quad (5.1)$$

$$\langle R, S \rangle \triangleq R \pi_1^\circ \cap S \pi_2^\circ \quad c \in \langle R, S \rangle \iff c R a \wedge c S b. \quad (5.2a)$$

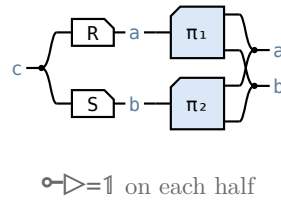
$$\langle R, S \rangle \pi_1 = \text{dom}(S) R \quad (5.6)$$

$$\langle R, S \rangle \pi_2 = \text{dom}(R) S \quad (5.7)$$



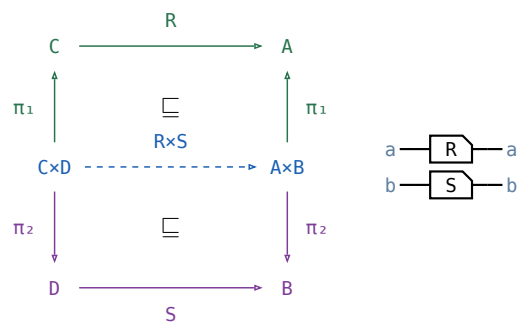
A domain is coreflexive, so $\langle R, S \rangle \pi_1 \sqsubseteq R$, with equality exactly when S is entire; for maps both triangles commute and $\langle f, g \rangle$ is unique. In $\mathbf{Rel}$, $c \in \langle R, S \rangle \iff c R a \text{ and } c S b$ — copy c , then R on one strand and S on the other, which is $\triangleleft(R \otimes S)$ on the right.

No $^\circ$ survives the translation. $\pi_1 = 1 \otimes \bullet$ discards the second component, so $\pi_1^\circ = 1 \otimes \bullet$ **creates** one out of nothing, and $\cap$ is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:



§5.2.1. Relational product $R \times S$

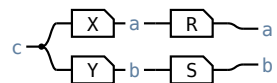
$$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle \text{ — the relational product } R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle. \quad (5.2)$$



Right-then-up is $(R \times S) \pi_1$, up-then-right is $\pi_1 R$, and $(R \times S) \pi_1 \sqsubseteq \pi_1 R$, equality when S is entire. In $\mathbf{Rel}$, $(c, d) \in (R \times S) \iff c R a \text{ and } d S b$ — two strands side by side, no copy dot: $R \times S = R \otimes S$.

§5.2.2. Absorption

For $X : E \rightarrow C$ and $Y : E \rightarrow D$, $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$. Both sides are this picture:



(5.2.2a)

	the law	picture
1	$\langle X, Y \rangle \langle R, S \rangle^\circ = (X \times R^\circ) \cap (Y \times S^\circ)$	
2	$\langle R, S \rangle^\circ \langle U, V \rangle \subseteq (R^\circ \times U) \times (S^\circ \times V)$	
3	$\text{Map}(f) \Rightarrow f \langle R, S \rangle = \langle f \times R, f \times S \rangle$ f a map; it fails for an arbitrary arrow;	
4	$X \langle R, S \rangle \subseteq (X \times R, X \times S)$ the half of the row above that holds for every X	
5	$F(R \times S) \text{ unzip}(F) \subseteq \text{unzip}(F) (F(R) \times F(S))$ $\text{unzip}(F) \triangleq \langle F(\pi_1), F(\pi_2) \rangle$; only $\subseteq$ for an arbitrary R, S	
6	$g = \text{curry}(f) \iff (g \times 1) \text{ eval} = f$ reading $\times$ as the relational product, does $\mathbf{Rel}$ have exponentials?	

§5.3. Relational coproducts

(5.3a)

the statement	picture
$R : a_1 \rightarrow c, S : a_2 \rightarrow c$	
1 $[R, S] \triangleq (l^\circ R) \cup (r^\circ S)$ (5.9) The tape is the union — a particle entering at A+B takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint.	
2 $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$	
3 $l [R, S] = R, r [R, S] = S$ $l T = R \wedge r T = S \Rightarrow T = [R, S],$	
4 $U : a_1 \rightarrow D, V : a_2 \rightarrow D$ $[U, V]^\circ [R, S] = (U^\circ R) \cup (V^\circ S) \quad (5.11)$	
5 $Z : c \rightarrow D$ $[R, S] Z = [R Z, S Z]$	

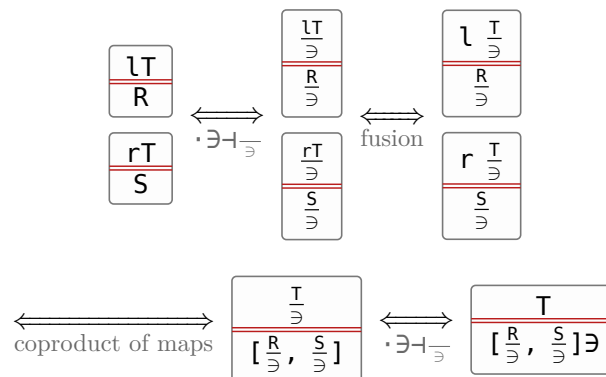
(5.3b)

the statement	picture
$R : a_1 \rightarrow b_1, S : a_2 \rightarrow b_2$	
1 $R + S \triangleq [R l, S r] \quad (5.10)$	
2 $l l^\circ = 1, r r^\circ = 1 \quad \text{Ex 5.12}$	$a_1 \xrightarrow{l} a \xrightarrow{l^\circ} a_1 = a_1 \xrightarrow{a} a_1 \quad a_2 \xrightarrow{r} a \xrightarrow{r^\circ} a_2 = a_2 \xrightarrow{a} a_2$
3 $l r^\circ = 0, r l^\circ = 0 \quad \text{Ex 5.12}$	$a_1 \xrightarrow{l} a \xrightarrow{r^\circ} a_2 = a_1 \xrightarrow{0} a_2$
4 $(l^\circ l) \cup (r^\circ r) = 1 \quad \text{Ex 5.12}$	

§5.3.1. $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$ — a case of relations is the case of their transposes, then $\exists$

The universal-property row is not free: l, r were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because $\exists$ sends an arrow $A \rightarrow C$ to a map $A \rightarrow PC$ reversibly, so the map coproduct can be applied underneath it. For any $T : A+B \rightarrow C$,

(5.3.1a)



(5.3.1b)



Nothing here holds only up to $\cong$: every triangle commutes on the nose, which is the difference from the fork above. The border spells $[R, S] = [\frac{R}{\bar{l}}, \frac{S}{\bar{r}}] \exists$, and pushing $\exists$ into the union that $[\cdot, \cdot]$ on maps already is turns that back into the definition, $(\bar{l} \circ \frac{R}{\bar{l}} \cup \bar{r} \circ \frac{S}{\bar{r}}) \exists = \bar{l} \circ R \cup \bar{r} \circ S$.

(5.3.1c)

the law	picture
1 prove (5.11), and say why duality does not carry it over from the product law	
2 $(R + S) \cap ([P, Q] [U, V]^\circ)$ $= (R \cap (P \cup^\circ)) + (S \cap (Q \cup^\circ))$ $[P, Q] [U, V]^\circ$ is a full 2×2 of composites; $R+S$ is diagonal, so the meet cuts the two off-diagonal branches	

§5.4. The power relator $P(R)$

(5.4a)

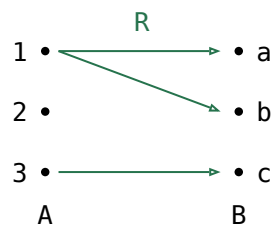
$P(R) \triangleq (\epsilon \setminus (R \epsilon)) \cap ((\exists R) / \exists)$ — the power relator $P(R) : PA \rightarrow PB$, R lifted to sets.

$xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$

(5.4b)

	in Rel	in words	$\text{im}(xs)$	partner
1	$(\exists R) / \exists$	$\forall y \in ys. \exists x \in xs. x \ R \ y$	$ys \subseteq \text{im}(xs)$	every y has a partner
2	$(\exists / (\exists R))^\circ$	$\forall y. (\exists x \in xs. x \ R \ y) \rightarrow y \in ys$	$ys \supseteq \text{im}(xs)$	ys leaves out no partner
3	$E(R)$	$ys = \{y \mid \exists x \in xs. x \ R \ y\}$	$ys = \text{im}(xs)$	every y has a partner, and none is left out
4	$((\exists R^\circ) / \exists)^\circ$	$\forall x \in xs. \exists y \in ys. x \ R \ y$	—	every x has a partner
5	$P(R)$	$P(R)(xs, ys) \iff (\forall y. y \in ys \implies (\exists x. x \in xs \wedge R(x, y))) \wedge (\forall x. x \in xs \implies (\exists y. y \in ys \wedge R(x, y)))$	—	every x and every y has a partner

(5.4c)



	xs	$(\exists R) / \exists$	$(\exists / (\exists R))^\circ$	$E(R)$	$((\exists R^\circ) / \exists)^\circ$	$P(R)$
1	$\emptyset$	$\emptyset$	all 8 subsets of abc	$\emptyset$	all 8 subsets of abc	$\emptyset$
2	$\{1\}$	$\emptyset, a, b, ab$	ab, abc	ab	a, b, ab, ac, bc, abc	a, b, ab
3	$\{2\}$	$\emptyset$	all 8 subsets of abc	$\emptyset$	none	none
4	$\{3\}$	$\emptyset, c$	c, ac, bc, abc	c	c, ac, bc, abc	c
5	$\{1, 2\}$	$\emptyset, a, b, ab$	ab, abc	ab	none	none
6	$\{1, 3\}$	all 8 subsets of abc	abc	abc	ac, bc, abc	ac, bc, abc
7	$\{2, 3\}$	$\emptyset, c$	c, ac, bc, abc	c	none	none
8	$\{1, 2, 3\}$	all 8 subsets of abc	abc	abc	none	none

(5.4d)

	law	the reading
1	$X \sqsubseteq P(R) \iff X \exists \sqsubseteq \exists R \wedge X^\circ \exists \sqsubseteq \exists R^\circ$	One containment, and the same one at R° — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$, and $R \sqsubseteq S \implies P(R) \sqsubseteq P(S)$.
2	$P(1) = 1$	The straightness axiom verbatim: extensionality is P 's unit law.
3	$\text{Map}(f) \implies P(f) = E(f)$	In Rel, $xs \ P(f) \ ys \iff ys = \{f(a) \mid a \in xs\}$. The half at f° says every $a \in xs$ has its $f(a)$ on ys ; f has just the one image per a , so that already says ys contains everything xs reaches, which is the fraction's second half. For a map the two definitions coincide.
4	$P(R \ S) = P(R) \ P(S)$	$\exists$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(RS)$.

§5.4.1. Type relators

(5.4.1a)

$T(R) = \langle F(R, 1) \ \alpha \rangle$ — the type functor T acts on an arrow $R : A \rightarrow B$ by rebuilding with α , applying R to the parameter.

(5.4.1b)

	name	law	what it says
1	the defining equation	$T(R) = \langle F(R, 1) \ \alpha \rangle$	Rebuild the structure with α , applying R to the parameter on the way.
2	type relator	$T(R)^\circ = T(R^\circ)$	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

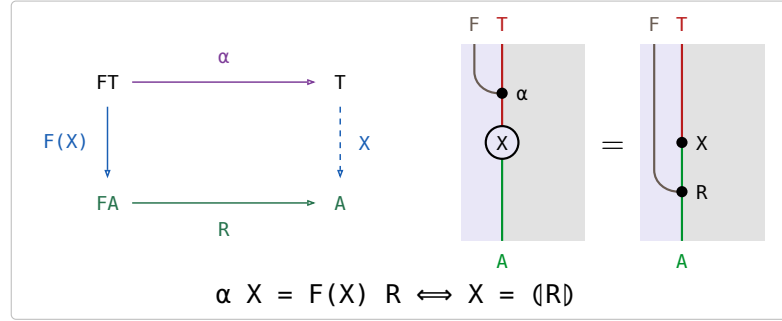
§5.5. Relational catamorphisms

(5.5a)

The initial algebra $\alpha : F(T) \rightarrow T$ among the maps is also initial in the allegory:

§5.5.1. The defining equation

(5.5.1a)



An F-algebra is a **weakened** natural transformation. A transformation $F \Rightarrow 1$ on $\mathcal{A}$ would need a component $FX \rightarrow X$ at every object and a commuting square at every arrow, but F-Algebra only need it works on T and A .

(5.5.1b)

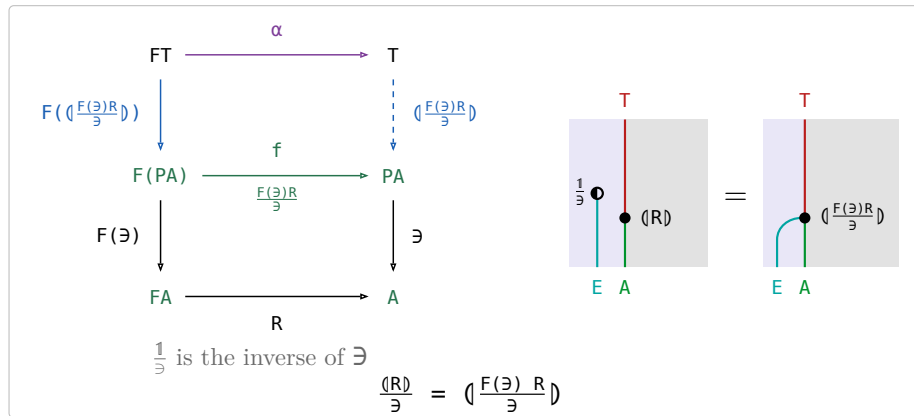
$$\langle [c, f] \rangle = \text{reduce}(c, f)$$

catamorphism: arrows of the form $\langle f \rangle$ are called catamorphisms, and these two examples reveal the notion to be a familiar idea in abstract clothing

1	part	Nat	[A]
2	datatype	$\text{Nat} ::= \text{zero} \mid \text{succ Nat}$	$[A] ::= \text{nil} \mid \text{cons}(A, [A])$
3	base functor F	$F(X) = 1 + X$	$F(X) = 1 + A \times X$
4	initial algebra α	$\alpha = [\text{zero}, \text{succ}]$ $: 1 + \text{Nat} \rightarrow \text{Nat}$	$\alpha = [\text{nil}, \text{cons}]$ $: 1 + A \times [A] \rightarrow [A]$
5	the fold, pointwise	$\langle [c, f] \rangle(\text{zero}) = c$ $\langle [c, f] \rangle(\text{succ}(n)) = f(\langle [c, f] \rangle(n))$	$\langle [c, f] \rangle(\text{nil}) = c$ $\langle [c, f] \rangle(\text{cons}(a, x)) = f(a, \langle [c, f] \rangle(x))$

§5.5.2. $\langle R \rangle = \langle \frac{F(\exists) R}{\exists} \rangle \exists$ — folding with R is a fold of maps into sets, then $\exists$

(5.5.2a)



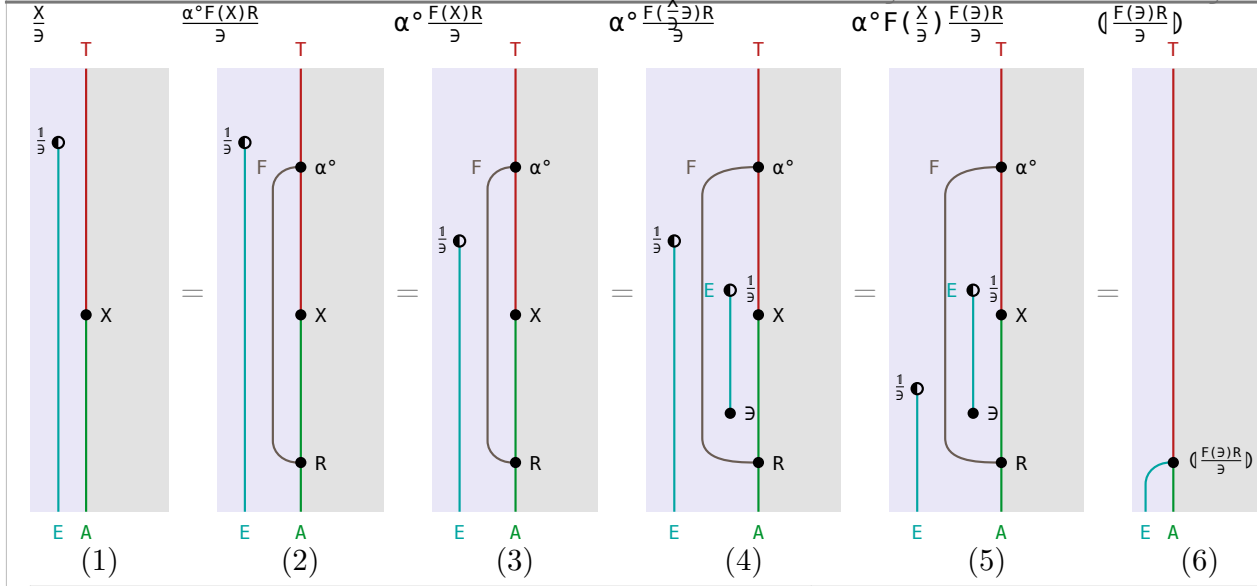
(5.12)

$$\begin{array}{c}
 \boxed{\frac{\alpha X}{F(X)R}} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \boxed{\frac{\frac{\alpha X}{\exists}}{\frac{F(X)R}{\exists}}} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \boxed{\frac{\frac{\alpha X}{\exists}}{\frac{F(\frac{X}{\exists})R}{\exists}}} \\
 \xleftrightarrow{\text{relator, fusion twice}} \boxed{\frac{\frac{\alpha}{\frac{X}{\exists}}}{\frac{F(\frac{X}{\exists})}{\frac{F(\exists)R}{\exists}}}} \xleftrightarrow{\text{reduce of maps}} \boxed{\frac{\frac{X}{\exists}}{\langle \frac{F(\exists)R}{\exists} \rangle}} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \boxed{\frac{X}{\langle \frac{F(\exists)R}{\exists} \rangle \exists}}
 \end{array}$$

the last two rows at $X := \langle R \rangle$:

$$\alpha X = F(X) R \iff \frac{X}{\exists} = \langle \frac{F(\exists) R}{\exists} \rangle \quad (5.5.2b)$$

in a tabular allegory, with F a relator and $\alpha : F(T) \rightarrow T$ its initial algebra: a relation X out of T satisfies the fold equation $\alpha X = F(X) R$ of R exactly when $\frac{X}{\exists}$ is the fold of the map $\frac{F(\exists) R}{\exists}$



1	$\frac{X}{\exists}$	α iso; $\alpha X = F(X) R$
2	$= \frac{\alpha^o F(X) R}{\exists}$	α^o is a map because α is an iso, so it leaves Λ : $\frac{\alpha^o F(X) R}{\exists} = \alpha^o \frac{F(X) R}{\exists}$ — $\text{Map}(f)$ $\implies \frac{f R}{\exists} = f \frac{R}{\exists}$
3	$= \alpha^o \frac{F(X) R}{\exists}$	$X = \frac{X}{\exists} \frac{R}{\exists} \implies \frac{R}{\exists} \exists = R$
4	$= \alpha^o \frac{F(\frac{X}{\exists}) R}{\exists}$	$F(\frac{X}{\exists}) = F(\frac{X}{\exists}) F(\exists)$ as F is a functor, and $F(\frac{X}{\exists})$ is a map (a relator sends maps to maps), so it leaves Λ — $\triangleq$ relator, $\text{Map}(f)$ $\implies \frac{f R}{\exists} = f \frac{R}{\exists}$
5	$= \alpha^o F(\frac{X}{\exists}) \frac{F(\exists) R}{\exists}$	fold uniqueness, $\frac{F(\exists) R}{\exists}$ being a map — $(h : f \rightarrow g) \triangleq f h = F(h) g$, $\text{Map}(f)$ $\implies \alpha \langle f \rangle = F(\langle f \rangle) f$
6	$= \langle \frac{F(\exists) R}{\exists} \rangle$	

§5.6. Combinatorial functions

(5.6a)

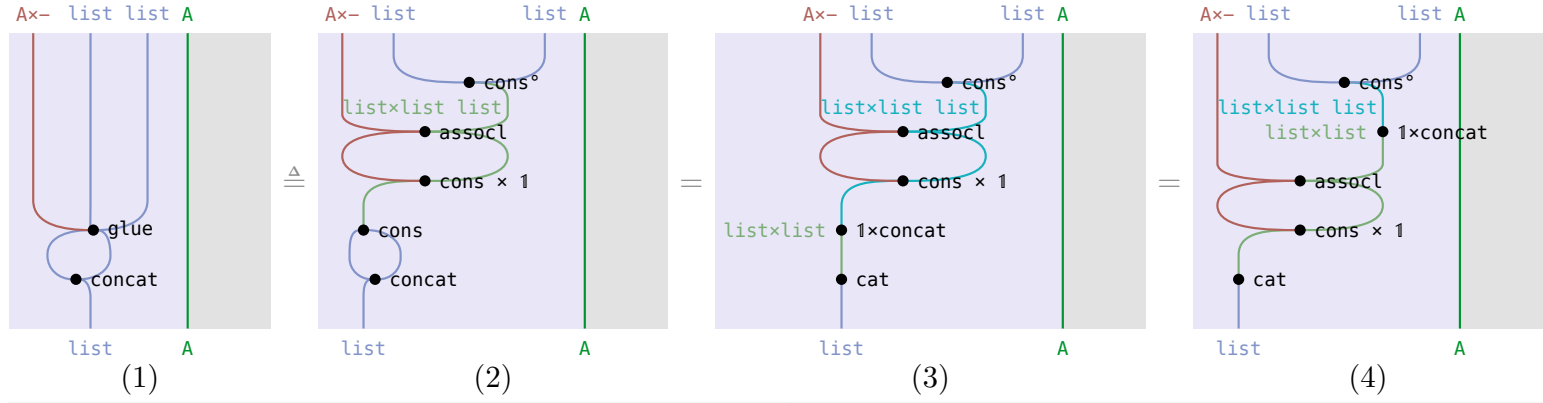
	definition	type	note
1	$[A] ::= \text{nil} \mid \text{cons}(A, [A]) \quad \text{list}(A) = [A]$	$\mathcal{A} \rightarrow \mathcal{A}$	The list type, under the short name it keeps.
2	$\text{list}(R) = \langle [\text{nil}, (R \times 1) \text{ cons}] \rangle$	$[A] \rightarrow [B]$	The relator's action on $R : A \rightarrow B$: one R per element, the shape untouched.
3	$\text{subseq} = \langle [\text{nil}, \text{cons} \cup \pi_2] \rangle$	$[A] \rightarrow [A]$	$xs \text{ subseq } ys$: ys is xs with elements dropped — cons keeps the head, π_2 drops it.
4	$\text{prefix} = \langle [\text{nil}, \text{cons} \cup \rightarrow \text{nil}] \rangle$ $\text{prefix} = \text{cat}^\circ \pi_1 = \text{init}^*$	$[A] \rightarrow [A]$	ys is an initial segment of xs ; the first nil is where it stops early. $\text{init} \triangleq \text{snoc}^\circ \pi_1$.
5	$\text{suffix} = \text{cat}^\circ \pi_2 = \text{tail}^*$	$[A] \rightarrow [A]$	The dual, $\text{tail} \triangleq \text{cons}^\circ \pi_2$; as a reduce it needs snoc -lists.
6	$\text{partition} = \text{concat}^\circ$	$[A] \rightarrow [[A]]$	This cat is restricted to $[A]^+ \times [A] \rightarrow [A]$, so ys is a list of non-empty segments of xs .
7	$\text{concat} = \langle [\text{nil}, \text{cat}] \rangle$	$[[A]] \rightarrow [A]$	Joins the segments back up, which is why its converse splits a list.
8	inits	$[A] \rightarrow [[A]]$	Implements $\frac{\text{prefix}}{\exists}$, listing the prefixes by increasing length.
9	tails	$[A] \rightarrow [[A]]$	Implements $\frac{\text{suffix}}{\exists}$ by decreasing length — the opposite order.
10	$\text{cpp}, \text{cpl}, \text{cpr}$	$[A] \times [B] \rightarrow [A \times B]$ $[A] \times B \rightarrow [A \times B]$ $A \times [B] \rightarrow [A \times B]$	Every pair (a, b) with a from the first list and b from the second; cpl , cpr keep one side a single element.
11	$\text{cpp setify} = (\text{setify} \times \text{setify}) \frac{\exists \times \exists}{\exists}$ $\text{cpl setify} = (\text{setify} \times 1) \frac{\exists \times 1}{\exists}$ $\text{cpr setify} = (1 \times \text{setify}) \frac{1 \times \exists}{\exists}$	$[A] \times [B] \rightarrow P(A \times B)$	Forgetting the order with setify , cpp is the transpose of $\exists \times \exists$.
12	$\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$	$F(PA) \rightarrow PFA$	$\text{cp}(F)$, the transpose of a relator's action on $\exists$; cpp implements it at $F(A) = A \times A$.
13	$\text{cp}(\text{list}) = \langle \frac{[\text{nil}, (\exists \times \exists) \text{ cons}]}{\exists} \rangle$ $\frac{[\text{nil}, (\exists \times \exists) \text{ cons}]}{\exists} = [\text{nil} \frac{1}{\exists}, \frac{\exists \times \exists}{\exists}]$ $E(\text{cons})]$	$\text{list}(PA) \rightarrow P(\text{list}(A))$	$\text{cp}(\text{list})$: (5.12) $\langle R \rangle = \langle \frac{F(\exists)}{\exists} R \rangle \exists$ at $\text{list}(\exists) = \langle [\text{nil}, (\exists \times 1) \text{ cons}] \rangle$, then the algebra expanded.
14	$\text{cplist} = \langle [\text{nil} \text{ wrap}, \text{cpp list}(\text{cons})] \rangle$	$[[A]] \rightarrow [[A]]$	$\text{cp}(\text{list})$ with every set a list: cpp in place of $\frac{\exists \times \exists}{\exists}$.

glue concat $\sqsubseteq (1 \times \text{concat}) \text{ cons}$

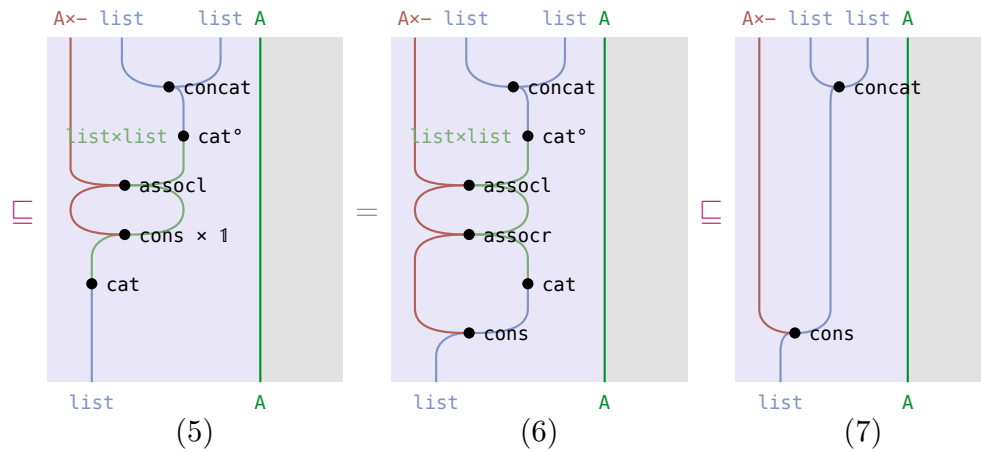
(5.6b)

putting **a** on the front of the first segment and flattening (**glue concat**) gives at most flattening the segments and putting **a** on the front of the result (**(1×concat) cons**)

the starting composite



1	glue concat	definition of glue
2	$\triangleq (1 \times \text{cons}^\circ) \text{ assocl } \text{cons} \times 1 \text{ cons } \text{concat}$	since $\text{concat} = ([\text{nil}, \text{cat}])$
3	$= (1 \times \text{cons}^\circ) \text{ assocl } \text{cons} \times 1 (1 \times \text{concat}) \text{ cat}$	naturality of assocl
4	$= (1 \times (\text{cons}^\circ (1 \times \text{concat}))) \text{ assocl } \text{cons} \times 1 \text{ cat}$	since $\text{concat} = ([\text{nil}, \text{cat}])$



5	$\sqsubseteq (1 \times \text{concat}) (1 \times \text{cat}^\circ) \text{ assocl } \text{cons} \times 1 \text{ cat}$	since $(\text{cons} \times 1) \text{ cat} = \text{assocr } (1 \times \text{cat}) \text{ cons}$
6	$= (1 \times \text{concat}) (1 \times \text{cat}^\circ) \text{ assocl } \text{assocr } (1 \times \text{cat}) \text{ cons}$	since $\text{assocl } \text{assocr} = 1$ and $\text{cat}^\circ \text{ cat} \sqsubseteq 1$
7	$\sqsubseteq (1 \times \text{concat}) \text{ cons}$	

§5.6.1. $\frac{\text{subseq}}{\exists} = ([\text{nil } \frac{1}{\exists}, (\frac{1 \times \exists}{\exists} \text{ E}(\text{cons}), \pi_2) \text{ cup}])$ — the set of subsequences is a fold of maps into sets

(5.6.1a)

$\text{cup} \triangleq \frac{(\pi_1 \exists) \cup (\pi_2 \exists)}{\exists}$ — cup unions two sets.

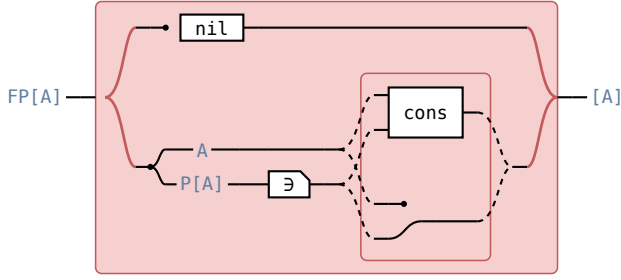
$\text{cup} \triangleq \frac{\pi_1 \exists \cup \pi_2 \exists}{\exists} : \text{PA} \times \text{PA} \rightarrow \text{PA}$

$$\frac{F(\exists) [\text{nil}, \text{cons } u \pi_2]}{\exists} = [\text{nil } \frac{1}{\exists}, \frac{(1 \times \exists) (\text{cons } u \pi_2)}{\exists}] \quad (5.6.1b)$$

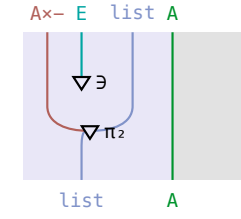
the set of lists the algebra builds is, from nothing, just **nil**, and from a head and a set of tails, every tail in the set with the head put on or left off — (5.12) $\langle R \rangle = \langle \frac{F(\exists)}{\exists} R \rangle \exists$ at **subseq** = $\langle [\text{nil}, \text{cons } u \pi_2] \rangle$, **subseq** = $\langle [\text{nil}, \text{cons } u \pi_2] \rangle$.

circuit — the fork is $F([A])=1+A \times [A]: \text{nil}$ above, the pair below

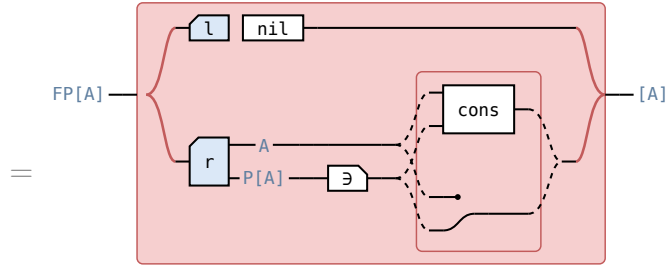
Hinze–Marsden



$$\frac{F(\exists) [\text{nil}, \text{cons } u \pi_2]}{\exists}$$

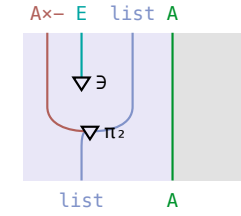


the π_2 operand of $\text{cons } u \pi_2$ under the $1 \times \exists$ summand of $F(\exists)$, i.e. $(1 \times \exists) \pi_2$

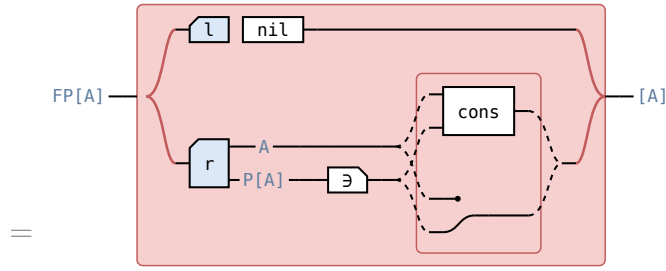


$$\frac{(1+1 \times \exists) [\text{nil}, \text{cons } u \pi_2]}{\exists}$$

$$F(X)=1+A \times X \text{ — list}(A) = [A]$$

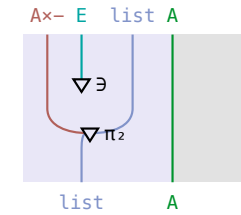


the same operand under $1+1 \times \exists$, whose $1 \times \exists$ summand it sits in

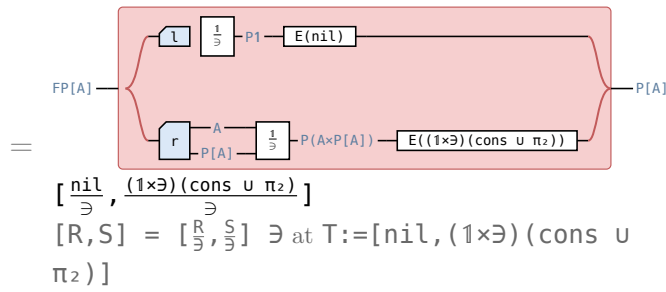


$$\frac{[\text{nil}, (1 \times \exists) (\text{cons } u \pi_2)]}{\exists}$$

$$R+S \triangleq [Rl, Sr], l[R, S]=R, r[R, S]=S \text{ — } [R, S] \triangleq (l^\circ R) \cup (r^\circ S), l[R, S]=R, r[R, S]=S, l T=R \wedge r T=S \Rightarrow T=[R, S]$$

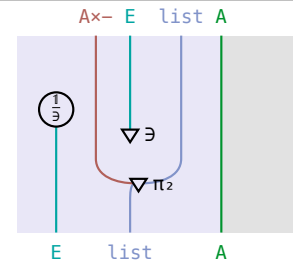


the π_2 operand of the second arm $(1 \times \exists) (\text{cons } u \pi_2)$

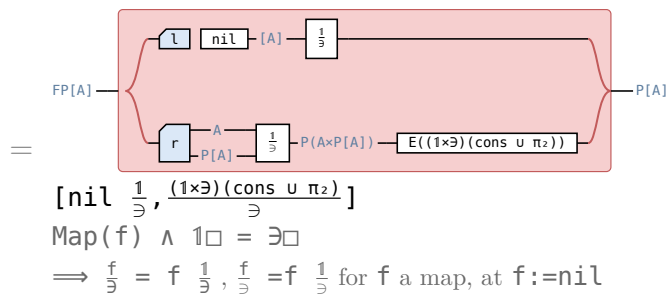


$$\frac{[\text{nil}, (1 \times \exists) (\text{cons } u \pi_2)]}{\exists}$$

$$[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists \text{ at } T := [\text{nil}, (1 \times \exists) (\text{cons } u \pi_2)]$$



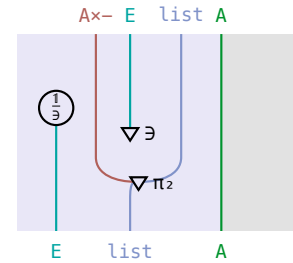
the π_2 operand under its $\frac{1}{\exists}$, the arm $(1 \times \exists) \pi_2 = \pi_2 \exists$ rewrites



$$\frac{[\text{nil } \frac{1}{\exists}, (1 \times \exists) (\text{cons } u \pi_2)]}{\exists}$$

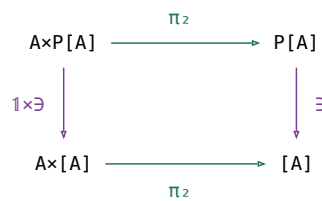
$$\text{Map}(f) \wedge 1 \square = \exists \square$$

$$\Rightarrow \frac{f}{\exists} = f \frac{1}{\exists}, \frac{f}{\exists} = f \frac{1}{\exists} \text{ for } f \text{ a map, at } f := \text{nil}$$



the same operand; the two rows differ only in the **nil** arm

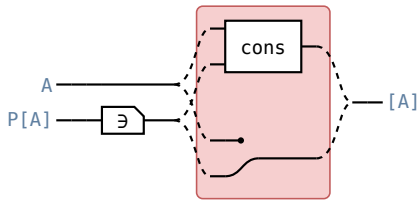
(5.6.1c)



$$\frac{(1 \times \exists) (\text{cons} \cup \pi_2)}{\exists} = \left(\frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup} \quad (5.6.1d)$$

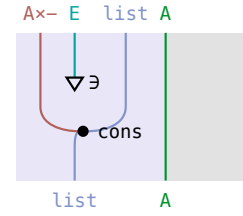
power transpose of join: the power transpose of the join of two relations is $\left(\frac{R}{\exists}, \frac{S}{\exists} \right) \text{cup}$, where **cup** is the function that returns the union of two sets

circuit

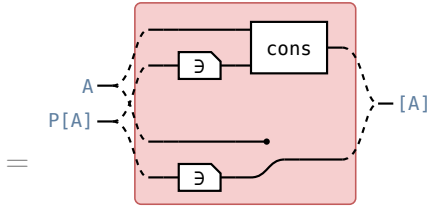


$$\frac{(1 \times \exists) (\text{cons} \cup \pi_2)}{\exists} = \frac{F(\exists) [\text{nil}, \text{cons} \cup \pi_2]}{\exists} = [\text{nil} \frac{1}{\exists}, \frac{(1 \times \exists) (\text{cons} \cup \pi_2)}{\exists}] \text{'s second branch}$$

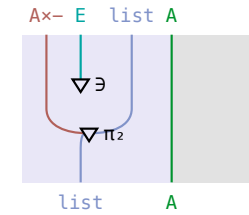
Hinze–Marsden



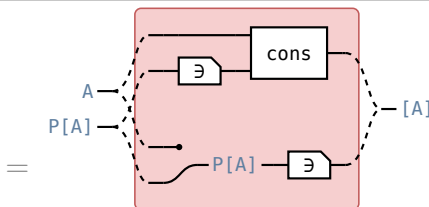
the **cons** operand of **cons** $\cup$ π_2



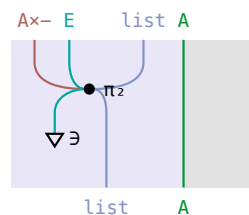
$$\frac{(1 \times \exists) \text{cons} \cup (1 \times \exists) \pi_2}{\exists} = T(X_1 \cup X_2) = TX_1 \cup TX_2 \text{ — } (S \cup T) R = (S R) \cup (T R), R(S \cup T) = (R S) \cup (R T), R \cap (S \cup T) = R \cap S \cup R \cap T, (R \cup S)^\circ = S^\circ \cup R^\circ$$



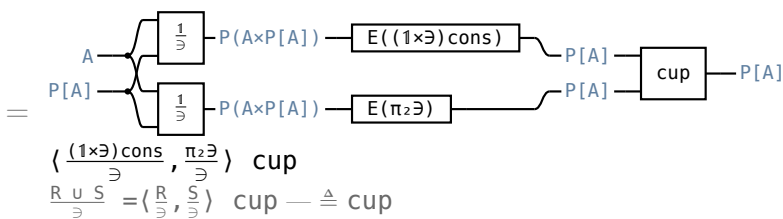
the π_2 operand of $(1 \times \exists) \text{cons} \cup (1 \times \exists) \pi_2$



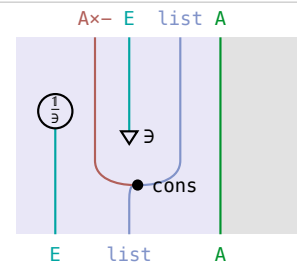
$$\frac{(1 \times \exists) \text{cons} \cup \pi_2 \exists}{\exists} = (1 \times \exists) \pi_2 = \pi_2 \exists \text{ — } (R \times S) \pi_1 \sqsubseteq \pi_1 R \wedge (R \times S) \pi_2 \sqsubseteq \pi_2 S \text{ at } \pi_2, \text{ an equality because } 1 \text{ is entire}$$



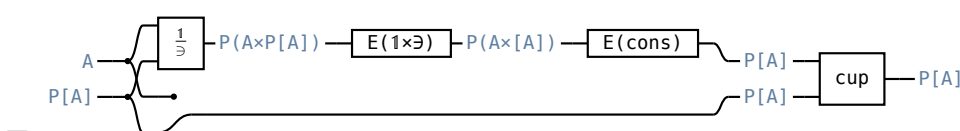
the π_2 operand of $(1 \times \exists) \text{cons} \cup \pi_2 \exists$



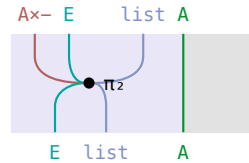
$$\left(\frac{(1 \times \exists) \text{cons}}{\exists}, \frac{\pi_2 \exists}{\exists} \right) \text{cup} = \frac{R \cup S}{\exists} = \left(\frac{R}{\exists}, \frac{S}{\exists} \right) \text{cup} \text{ — } \triangleq \text{cup}$$



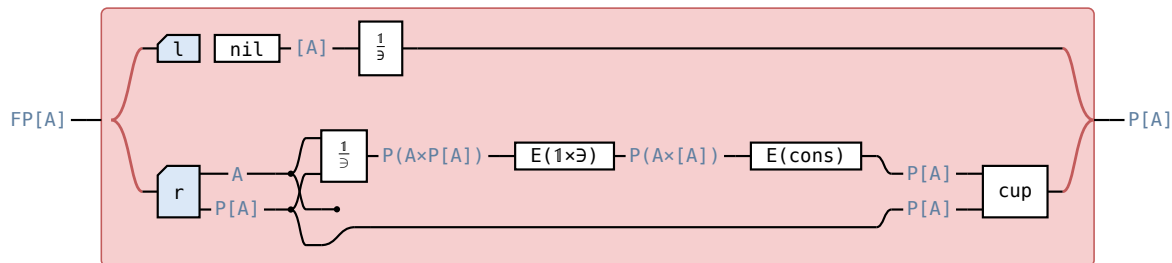
the **cons** operand under its $\frac{1}{\exists}$



$$\left(\frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup} = \frac{S}{\exists} E(R) = \frac{S}{\exists} R, \text{ absorption } \frac{S}{\exists} E(R) = \frac{SR}{\exists} \text{ at } S := 1 \times \exists, R := \text{cons}; \text{ fusion and } \frac{\exists}{\exists} = 1 \text{ on the } \pi_2 \text{ operand}$$



the π_2 operand, bare π_2



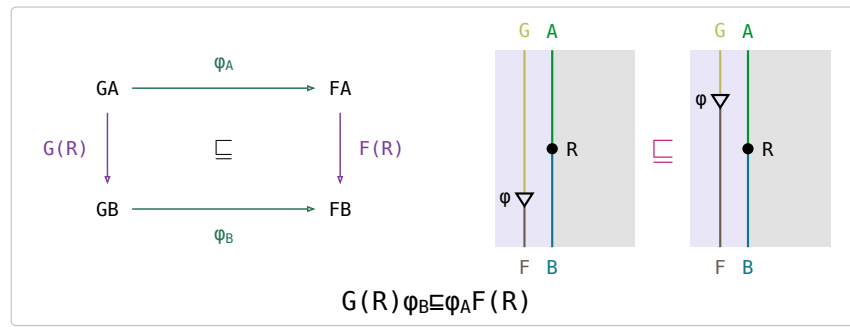
$$\left[\frac{\text{nil}}{\exists}, \left(\frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup} \right]$$

which writes $P\text{cons}$; **cons** is a map, and there $P(\text{cons}) = E(\text{cons}) \text{ — Map}(f) \Rightarrow P(f) = E(f)$.

(5.6.1e)

§5.7. Lax natural transformations (LaT)

$\varphi : G \Rightarrow F$ is **lax** at $R : A \rightarrow B$ when $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$; it is a **lax natural transformation** (LaT) when lax at every R . (5.7a)



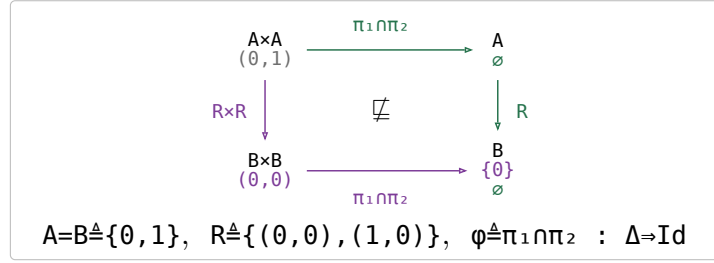
(5.7c)

	closed under	commutative diagram	string diagram
1	composition $\psi\phi$	$H(R)\psi_B \sqsubseteq \psi_A G(R)$ and $G(R)\phi_B \sqsubseteq \phi_A F(R)$ give $H(R)(\psi_B \phi_B) \sqsubseteq (\psi_A \phi_A) F(R)$	
2	horizontal composition $\chi \circ \phi$	$\phi : G \Rightarrow F$ with $G, F : \mathcal{C} \rightarrow \mathcal{D}$ and $\chi : L \Rightarrow K$ with $L, K : \mathcal{D} \rightarrow \mathcal{E}$ give $\chi \circ \phi : L \circ G \Rightarrow K \circ F$, the family $A \mapsto \chi_{GA} K(\phi_A)$ $L(G(R)) \chi_{GB} \sqsubseteq \chi_{GA} K(G(R))$ is χ lax at $G(R)$; then $K(G(R)\phi_B) \sqsubseteq K(\phi_A F(R))$ is K applied to ϕ's own inequation; $\chi := \mathbb{1}_K$ gives $K(\phi) : K \circ G \Rightarrow K \circ F$, K composed on the outside; $\phi := \mathbb{1}_G$ gives $\chi G : L \circ G \Rightarrow K \circ G$, G composed on the inside two candidates, $\chi_{GA} K(\phi_A)$ and $L(\phi_A) \chi_{FA}$, ordered by $\sqsubseteq$; this row is the first	
3	union $\phi \cup \psi$	$G(R)\phi_B \sqsubseteq \phi_A F(R)$ and $G(R)\psi_B \sqsubseteq \psi_A F(R)$ give $G(R)(\phi_B \cup \psi_B) \sqsubseteq (\phi_A \cup \psi_A) F(R)$	
4	a relator K $K(\phi)$	$G(R)\phi_B \sqsubseteq \phi_A F(R)$ gives $K(G(R))K(\phi_B) \sqsubseteq K(\phi_A)K(F(R))$	
5	product $\phi \times \psi$	$\phi : G \Rightarrow F$ and $\psi : G' \Rightarrow F'$ give $\phi \times \psi : G \times G' \Rightarrow F \times F'$ $(R \times S)(U \times V) = (RU) \times (SV)$ and monotonicity in both slots — the row above's K applied to the inequation, run on a bifunctor; the fork is the derived case $\langle \phi, \psi \rangle = \triangleleft (\phi \times \psi)$, and its ONE cost is the lax copy law $R \triangleleft \sqsubseteq \triangleleft (R \times R) \rightarrow R \triangleleft \sqsubseteq \triangleleft (R \otimes R)$, $R \multimap \sqsubseteq \multimap$	
6	coproduct $\phi + \psi$	$\phi : G \Rightarrow F$ and $\psi : G' \Rightarrow F'$ give $\phi + \psi : G + G' \Rightarrow F + F'$ $(R + S)(U + V) = (RU) + (SV)$ and monotonicity in both slots; the co-fork is the derived case $[\phi, \psi] = (\phi + \psi) \nabla$, and ∇ costs nothing the step it would need is $(\phi_A F(R)) \cap (\psi_A F(R)) \sqsubseteq (\phi \cap \psi)_A F(R)$, the wrong direction of R $(S \cap T) \sqsubseteq (R S) \cap (R T)$ a counterexample: $aa = (\emptyset, 1) \wedge bb = (\emptyset, 0) \wedge (R \times R)(aa, bb) \wedge b_1 = \{\emptyset\} \wedge b_1 = E(\pi_1 \cap \pi_2)\{bb\} \wedge a_1 = \emptyset \wedge a_1 = E(\pi_1 \cap \pi_2)\{aa\} \wedge b_2 = \emptyset \wedge b_2 = E(R)(a_1)$ $\Rightarrow \neg((R \times R)(\pi_1 \cap \pi_2) \sqsubseteq (\pi_1 \cap \pi_2) R)$	
7	meet — fails $\phi \cap \psi$		
8	Each row is the general slide $R_A X \sqsubseteq X' R_B$ and $R_B Y \sqsubseteq Y' R_C$ give $R_A(XY) \sqsubseteq (X'Y')R_C$ at $R_A, R_B := G(R), F(R)$ and $X, X' := \phi_B, \phi_A$, quantified over R ; at $X' = X$ with R_A, R_B endorelations it is the monotonicity reading instead — the $\forall R$ sits outside the law, and is the only difference		

§5.7.1. meet is not closed in LaT

counterexample — $\varphi, \psi : \mathbf{G} \Rightarrow \mathbf{F}$ lax natural does NOT give $\varphi \eta \psi$ lax natural

(5.7.1a)



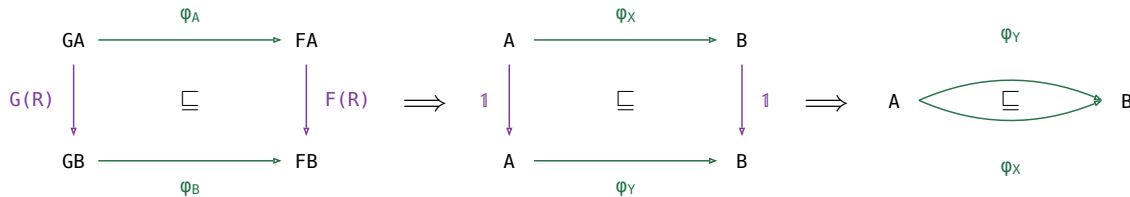
§5.7.2. Two stacked towers

(5.7.2a)

	lower — inside ONE allegory	upper — between allegories
1 0-cell	an object A	an allegory $\mathcal{C}$ — a whole lower tower
2 1-cell	a relation $R : A \rightarrow B$	a relator $F : \mathcal{C} \rightarrow \mathcal{D}$
3 2-cell	$R \sqsubseteq S$	a LaT $\varphi : \mathbf{G} \Rightarrow \mathbf{F} \triangleq \text{lax}$
4 order	$R \sqsubseteq S$	$\varphi \sqsubseteq \psi$ componentwise
5 union	$R \cup S$	$\varphi \cup \psi$, survives $\text{lax}(\varphi) \wedge \text{lax}(\psi) \Rightarrow G(R) (\varphi_B \cup \psi_B) \sqsubseteq (\varphi_A \cup \psi_A) F(R)$
6 product	$R \times S \triangleq \times$	$\varphi \times \psi$, survives; a TENSOR, not a categorical product; SYMMETRIC $\langle R, S \rangle$ $\pi_1 = \text{dom}(S) R, \langle R, S \rangle \pi_2 = \text{dom}(R) S$ $\text{lax}(\varphi) \wedge \text{lax}(\psi) \Rightarrow (G(R) \times G'(R)) (\varphi_B \times \psi_B) \sqsubseteq (\varphi_A \times \psi_A) (F(R) \times F'(R))$
7 coproduct	$R + S$	$\varphi + \psi$, survives; a BIPRODUCT $\text{lax}(\varphi) \wedge \text{lax}(\psi) \Rightarrow (G(R) + G'(R)) (\varphi_B + \psi_B) \sqsubseteq (\varphi_A + \psi_A) (F(R) + F'(R))$
8 meet	$R \sqcap S$	$\varphi \eta \psi$ componentwise, fails $aa = (0,1) \wedge bb = (0,0) \wedge (R \times R)(aa, bb) \wedge b_1 = \{0\} \wedge b_1 = E(\pi_1 \eta \pi_2)\{bb\} \wedge a_1 = \emptyset \wedge a_1 = E(\pi_1 \eta \pi_2)\{aa\} \wedge b_2 = \emptyset \wedge b_2 = E(R)(a_1) \Rightarrow \neg((R \times R) (\pi_1 \eta \pi_2) \sqsubseteq (\pi_1 \eta \pi_2) R)$
9 converse	R°	φ° , fails, oplax: $\varphi_A^\circ G(R) \sqsubseteq F(R) \varphi_B^\circ$
10 zero object	z with $1 \ z = \emptyset$	the constant relator at such a z , initial and terminal

§5.7.3. RES is a special case of LaT

(5.7.3a)

 $\mathbf{G} \triangleq \text{const } A, \mathbf{F} \triangleq \text{const } B$ — the relators $X \rightarrow A$ and $X \rightarrow B$, each sending every arrow to $1 \triangleq \text{lax}$ with $\mathcal{C}$ two objects and one non-identity arrow $X \rightarrow Y$, a LaT $\text{const } A \Rightarrow \text{const } B$ is the pair $\varphi_X, \varphi_Y : A \rightarrow B$ with $\varphi_Y \sqsubseteq \varphi_X$ — the datum $R \sqsubseteq S$ not every LaT is constant: $\exists : \mathbf{P} \Rightarrow \text{Id} \ X \sqsubseteq P(R) \iff X \exists \sqsubseteq \exists R \wedge X^\circ \exists \sqsubseteq \exists R^\circ \ \pi_2 : (A \times -) \Rightarrow \text{Id} \ (1 \times \exists) \pi_2 = \pi_2 \exists$ $\triangleleft : \text{Id} \Rightarrow \Delta \ R \triangleleft \leq \triangleleft (R \otimes R), R \multimap \leq \multimap$